



GETTING STARTED

1. CDN

```
<head>
<script src =
'https://cdn.plot.ly/plotly-latest.
min.js'></script>
</head>
```

2. Sign Up & Configure

plot.ly/javascript/getting-started

3. A Hello World Figure

```
JS:
<script>
myDiv =
document.getElementById (
'myDiv' );
data = { x : [ 1, 2, 3, 4, 5 ],
y : [ 1, 2, 4, 8, 16 ] };
trace = [ data ];
Plotly.plot ( myDiv , trace );
</script>

HTML:
<head>
<script src =
'https://cdn.plot.ly/plotly-latest.
min.js'></script>
</head>
<div id = 'myDiv' ></div>
```

BASIC CHARTS

Line Plots

```
trace1 = {
x : [ 1, 2 ], y : [ 1, 2 ]
type = 'scatter' };
trace2 = {
x : [ 3, 4 ], y : [ 9, 16 ]
type = 'scatter' };
Plotly.plot (
div , [ trace1 , trace2 ] );
```

Scatter Plots

```
trace = {
x : [ 1, 2, 3 ],
y : [ 1, 2, 3 ]
text : [ 'A', 'B', 'C' ]
textposition : 'top center'
mode : 'markers+text' };
Plotly.plot ( div , [ trace ] );
```

Bubble Charts

```
trace {
x : [ 1, 2, 3 ],
y : [ 1, 2, 3 ],
marker : {
color : [ 'red', 'blue' ],
size : [ 20, 50, 80 ] },
mode : 'markers' };
Plotly.plot (
'myDiv', [ trace ] );
```

Heatmaps

```
trace = {
z : [ [ 1, 2 ], [ 1, 2 ] ],
type : 'heatmap',
data : [ trace ],
Plotly.plot (
'myDiv', data );
```

Bar Charts

```
trace = {
x : [ 1, 2 ],
y : [ 1, 2 ]
type : 'bar' };
data = [ trace ];
Plotly.plot ( div , data );
```

Area Plots

```
trace {
x : [ 1, 2, 3 ],
y : [ 1, 2, 3 ],
type : 'scatter',
type : 'tonexty' };
Plotly.plot (
'myDiv', [ trace ] );
```

LAYOUT

Legends

```
trace1 = {
x : [ 1, 2, 3 ],
y : [ 1, 2, 3 ]
name = 'Calvin',
type = 'scatter' };
trace2 = {
x : [ 1, 2, 3 ],
y : [ 1, 2, 3 ]
name = 'Hobbes',
type = 'scatter' };
layout = {
showlegend = true ,
legend : {
x : 0.2,
y : 0.5 } };
```

Axes

```
trace = {
x : [ 1, 2, 3 ],
y : [ 1, 2, 3 ]
type = 'scatter' };
axis_template = {
showgrid = false ,
zeroline = false ,
nticks = 20 ,
showline = true ,
title = 'X Axis'
mirror = 'all' )
layout = {
axis : axis_template ,
yaxis = axis_template };
```

```
fig = {
data : [ trace ],
layout : layout };
Plotly.plot (
'myDiv', fig );
```

STATISTICAL CHARTS	
 Histograms	<pre>var trace = { x: [1, 2, 3, 4, 5], type = 'histogram' } Plotly.Plot('Div', [trace]);</pre>
 Box Plots	<pre>var trace = { y: [1, 2, 3, 4, 5], type = 'box' } Plotly.Plot('Div', [trace]);</pre>
 2D Histogram	<pre>var trace = { x: [1, 2, 3, 4, 5], y: [1, 2, 3, 4, 5], type = 'histogram2d' } Plotly.Plot('Div', [trace]);</pre>

MAPS	
 Bubble Map	<pre>trace = { type: 'scattergeo', lon: [100, 400], lat: [0, 0], marker: { color: ['red', 'blue'], size: [30, 50], mode = 'markers' } } Plotly.Plot('myDiv', [trace]);</pre>
 Choropleth Map	<pre>trace = { type = 'scattergeo', locations = ['AZ', 'CA', 'VT'], locationmode = 'USA-states', colorscale = 'Viridis', z = [10, 20, 40] }; layout = { geo: { scope = 'usa' } }; fig = { data: [trace], layout: layout }; Plotly.newPlot('myDiv', fig);</pre>
 Scatter Map	<pre>trace = { type = 'scattergeo', lon = [42, 39], lat = [12, 22], text = ['Rome', 'Greece'], mode = 'markers'; } Plotly.newPlot('myDiv', [trace]);</pre>

3D CHARTS	
 3D Surface Plots	<pre>trace = { colorscale = 'Viridis', z = [[3, 5, 7, 9], [[21, 13, 8, 5]]]; Plotly.newPlot('myDiv', [trace]);</pre>
 3D Line Plots	<pre>trace = { x = [9, 8, 5, 1], y = [1, 2, 4, 8], z = [11, 8, 15, 3], mode = 'lines' }; data = [trace]; Plotly.newPlot('myDiv', data);</pre>
 3D Scatter Plots	<pre>trace = { x = [9, 8, 5, 1], y = [1, 2, 4, 8], z = [11, 8, 15, 3], mode = 'markers' }; Plotly.newPlot('myDiv', [trace]);</pre>

FIGURE HIERARCHY	
Figure {}	DATA [] TRACE {} x, y, z [] color, text, size [] colorscale ABC or [] MARKER {} color ABC symbol ABC LINE {} color ABC width 123 LAYOUT {} title ABC showlegend True/False autosize True/False XAXIS, YAXIS {} SCENE {} XAXIS, VAXIS, ZAXIS {} GEO {} LEGEND {} ANNOTATIONS {} {} = dictionary [] = list ABC = string 123 = number